

Effectiveness of airport screening for Bundibugyo ebolavirus: assessing the risk of undetected importation to the United Kingdom

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Table of contents

Abstract	1
1 Introduction	2
2 Methods	3
2.1 Simulation model	3
2.2 Parameters	3
2.3 Uncertainty propagation	4
2.4 Sensitivity analyses	4
2.5 Software	5
3 Results	5
3.1 Detection outcomes at 12 hours	5
3.2 Effect of flight duration on detection probability	5
3.3 Detection outcome breakdown	5
3.4 Sensitivity analysis: asymptomatic proportion and incubation period	5
3.5 Sensitivity analysis: epidemic growth phase	10
3.6 Sensitivity analysis: screening sensitivity at exit and entry	12
4 Discussion	12
References	13

Abstract

We apply a stochastic simulation model to estimate the effectiveness of combined exit and entry airport screening for Bundibugyo ebolavirus (BDBV), using natural-history parameters from a Bayesian re-analysis of the 2012 Isiro outbreak. For a 12-hour DRC/Uganda-to-UK itinerary at 86% screening sensitivity, we estimate 65% of infected travellers would arrive undetected (95% CrI:

42–77%). The dominant mechanism of failure is the incubation period: most infected travellers board their flight before symptom onset and are undetectable by any fever-based screen. This is compounded during active epidemic growth, when recently exposed — and therefore pre-symptomatic — cases are overrepresented among travellers. Airport screening cannot provide robust containment of BDBV importation and should be considered a supplementary measure alongside clinician awareness and rapid response capacity.

i Note

Data note: Natural-history parameters (incubation period, onset-to-hospitalisation delay) are derived from the 2007 Uganda and 2012 DRC BDBV outbreaks. Patient-level data from the current 2026 DRC outbreak are not yet available. The epidemic doubling time range used in the growth-phase sensitivity analysis is estimated from the current outbreak (epiforecasts, 2026).

1 Introduction

Bundibugyo ebolavirus (BDBV) is one of six known ebolaviruses and was first identified during a 2007 outbreak in Bundibugyo district, Uganda, which caused 149 cases and 37 deaths (Towner et al. 2008). A second, larger outbreak occurred in Isiro, Haut-Uélé province, Democratic Republic of the Congo (DRC), between June and November 2012, resulting in 52 confirmed and probable cases and 28 deaths (Rosello et al. 2015). As of June 2026, an outbreak of BDBV in DRC has prompted renewed attention to the risk of international spread, including to high-income countries such as the United Kingdom, which maintain direct or connecting air links to the region.

Airport screening — comprising thermal scanning, symptom questionnaires, and direct clinical assessment — was deployed extensively during the 2014–2016 West Africa Ebola epidemic and again during the 2018–2020 DRC outbreaks of Zaire ebolavirus (Mabey, Flasche, and Edmunds 2014). Its effectiveness, however, is constrained by the natural history of infection: travellers who have been exposed but have not yet developed symptoms will not be febrile at the time of screening, and therefore will not be detected. Whether a traveller is symptomatic at departure or arrival depends on when they were exposed relative to their travel date, on the distribution of the incubation period (infection to symptom onset), and on the time from symptom onset to progression to severe, detectably ill disease.

For BDBV specifically, three published incubation period estimates exist. Wamala et al. (2010) (Wamala et al. 2010) reported a mean of 7.68 days (SD 3.53; 95% CI 7.04–8.32; $n = 116$) from the 2007 Uganda outbreak. MacNeil et al. (2010) (MacNeil et al. 2010) reported 6.3 days (95% CI 5.2–7.3; $n = 24$ contacts with confirmed exposure dates) from the same outbreak. Kratz et al. (2015) (Kratz et al. 2015) reported 11.3 days (range 2–27; $n = 3$) from the 2012 Isiro DRC outbreak. The Nash et al. (2024) systematic review (Nash et al. 2024) identified all three. The Isiro 2012 line list, the only publicly available BDBV line list, contains no exposure dates and was recently re-analysed using modern doubly-censored Bayesian methods to produce posterior distributions for the onset-to-admission and admission-to-death delays (Funk and Abbott 2026). We

use these posterior estimates directly, propagating their uncertainty into our screening effectiveness projections.

The primary route of concern is DRC or Uganda to the United Kingdom. Direct flights do not operate between these origins and the UK; travellers typically route via a European or Middle Eastern hub, giving a total travel time of approximately 12 hours. We model this default itinerary whilst also examining how screening effectiveness varies across a range of flight durations and epidemiological scenarios.

2 Methods

2.1 Simulation model

We used a stochastic individual-based simulation model to evaluate the probability that an infected traveller would be detected at (i) exit screening before departure, (ii) as severely ill during the flight, (iii) at entry screening on arrival, or (iv) remain undetected throughout. Full details of the model are given in Quilty et al. (2020) (Quilty et al. 2020); a brief summary follows.

For each simulated traveller, an incubation period t_{inc} and an onset-to-severe-disease interval t_{inf} were drawn independently from Gamma distributions. The time at which a traveller commenced their flight relative to their infection was drawn uniformly from $[0, 2(t_{\text{inc}} + t_{\text{inf}})]$, so that travellers at all stages of their natural history — from pre-symptomatic to late-symptomatic — are represented in proportion to the expected exposure distribution.

Exit screening was assumed to detect any traveller who was symptomatic ($t_{\text{flight departure}} > t_{\text{inc}}$) with probability p_{exit} . Travellers who were not symptomatic at departure but became severely ill before arrival (i.e. $t_{\text{flight departure}} \leq t_{\text{inc}}$ but $t_{\text{inc}} + t_{\text{inf}} < t_{\text{flight arrival}}$) were classified as severely ill during flight. Remaining symptomatic travellers — those who were symptomatic at arrival but not detected at exit — were subject to entry screening with probability p_{entry} .

Travellers who were hospitalised before their scheduled departure were removed from the simulated cohort, consistent with the assumption that severely ill individuals are unlikely to travel.

A proportion ρ of all infections was assumed to be asymptomatic; these travellers remained undetected throughout.

2.2 Parameters

Table 1: Model parameters for the BDBV default scenario.

Parameter	Value	Source
Flight duration	12 h (default)	DRC/Uganda to UK via hub
Exit screening sensitivity	86%	Priest et al. 2011 (thermal image scanners)
Entry screening sensitivity	86%	Priest et al. 2011 (thermal image scanners)
Proportion asymptomatic	5%	EVD literature (conservative)
Incubation period mean	7.68 days (95% CI: 7.04–8.32)	Wamala et al. 2010 (2007 Uganda BDBV; n=116)
Incubation period variance	12.46 days ² (SD ² = 3.53 ²)	Wamala et al. 2010
Between-estimate incubation SD	2.6 days (prior SD)	Sample SD across 3 BDBV estimates (Nash 2024 / ep

Onset → admission mean	4.03 days (95% CrI: 3.08–5.46)	Funk & Abbott 2026 posterior median
Onset → admission variance	13.72 days ²	Funk & Abbott 2026 posterior median

The onset-to-admission (i.e., onset to severe disease) parameters for BDBV were derived from the Bayesian re-analysis of the 2012 Isiro line list by Funk & Abbott (2026) (Funk and Abbott 2026), fitting Gamma distributions to doubly-interval-censored delay data. This analysis is available as a git submodule at `submodules/bdbv-linelist-analysis`. The model selected the Gamma family on the basis of WAIC (lowest WAIC = 792.4 versus 793.6 for Weibull and 652,838 for LogNormal). The posterior median mean was 4.0 days (95% CrI: 3.08–5.46) and the posterior median SD was 3.7 days (95% CrI: 2.66–5.58), corresponding to a Gamma shape of 1.18 and scale of 3.41.

Three published BDBV-specific incubation period estimates exist: Wamala et al. (2010) (Wamala et al. 2010) (mean 7.68 days, SD 3.53, 95% CI 7.04–8.32, $n = 116$, Uganda), MacNeil et al. (2010) (MacNeil et al. 2010) (mean 6.3 days, 95% CI 5.2–7.3, $n = 24$, Uganda), and Kratz et al. (2015) (Kratz et al. 2015) (median 11.3 days, range 2–27, $n = 3$, DRC). All three are identified in the Nash et al. (2024) systematic review (Nash et al. 2024). We used the Wamala estimate as the primary value owing to the largest sample size and direct SD estimate. The between-study standard deviation across the three estimates (sample SD 2.6 days) was used as the spread of the incubation prior. Uncertainty in the incubation period was propagated by sampling the incubation mean on each posterior draw from a Normal prior with mean 7.68 days and SD 2.6 days (truncated above 0.5 days), then scaling the variance to preserve the coefficient of variation.

2.3 Uncertainty propagation

For each of 1000 posterior draws sampled at random from `posterior_gamma.csv`, we ran the simulation with 2,000 simulated travellers. On each draw, the onset-to-admission delay was taken from the BDBV posterior (draw-specific Gamma mean and variance from the Funk & Abbott 2026 analysis), and the incubation period mean was drawn from a Normal prior with mean 7.68 days and SD 2.6 days (between-study spread across three BDBV estimates (Wamala et al. 2010; MacNeil et al. 2010; Kratz et al. 2015)), with variance rescaled to preserve the coefficient of variation.

2.4 Sensitivity analyses

Four sensitivity analyses were conducted. First, we examined how the proportion undetected varied across flight durations of 1 to 20 hours, with full posterior uncertainty. Second, we explored the joint effect of the proportion of asymptomatic infection (0 to 60%) and the mean incubation period (1 to 14 days) on the proportion undetected for the default 12-hour flight using posterior-median natural-history parameters. Third, to address the limitation that the baseline model assumes a stable epidemic, we re-ran the posterior analysis across a range of epidemic doubling times (4.5 to 44 days, spanning the 90% credible interval from the epiforecasts BDBV 2026 growth-rate estimates (Funk and Abbott 2026)) using the truncated-exponential exposure-time distribution described by Gostic et al. (K. M. Gostic, Kucharski, and Lloyd-Smith 2015; K. Gostic et al. 2020). Fourth, we varied exit and entry screening sensitivity independently from 0 to 100% to quantify the marginal contribution of each screening layer.

2.5 Software

All analyses were performed in R (version 4.1; R Core Team 2024). The simulation model is implemented in the `airportscreening` R package, available at github.com/bquilty25/airport_screening Ebola_bvd. Figures were produced using `ggplot2` (Wickham 2016). The BDBV parameter submodule uses Julia and Turing.jl.

3 Results

3.1 Detection outcomes at 12 hours

Table 2: Estimated screening outcomes for 100 infected BDBV travellers on a 12-hour flight from DRC/Uganda to the UK, with 86% sensitivity at exit and 86% sensitivity at entry. Values are percentages with 95% credible intervals.

Detection outcome	Estimate (95% CrI)
Detected at exit screening	26.3 (17.4–44.4)
Severely ill during flight	1.0 (0.3–2.6)
Detected at entry screening	7.1 (4.4–13.5)
Not detected	65.6 (41.9–76.6)

For a 12-hour DRC-or-Uganda-to-UK flight with 86% screening sensitivity at both departure and arrival, we estimate that 26.3% (95% CrI: 17.4–44.4%) of infected travellers would be detected at exit screening, 1.0% (95% CrI: 0.3–2.6%) would become severely ill during the flight, and 7.1% (95% CrI: 4.4–13.5%) would be detected at entry screening (Table 2). In total, 65.6% (95% CrI: 41.9–76.6%) of infected travellers would arrive undetected.

3.2 Effect of flight duration on detection probability

As flight duration increases, the probability of detecting an infected traveller at entry screening rises owing to progression of disease during travel (Figure 2). For the 12-hour DRC-to-UK scenario, combined exit and entry screening is estimated to detect 34.4% (95% CrI: 23.4–58.1%) of infected travellers, leaving 65.6% undetected. Even for a 20-hour flight — representative of the longest possible single-airline connection — the proportion undetected remains 61.6% (95% CrI: 38.4–75.9%).

3.3 Detection outcome breakdown

3.4 Sensitivity analysis: asymptomatic proportion and incubation period

The primary driver of the undetected fraction is the incubation period (Figure 4): even at the conservative BDBV default of 5% asymptomatic infection, the proportion undetected ranges from 15–79% across incubation periods of 1–14 days, with longer incubation periods allowing more travellers to depart before symptom onset. Truly asymptomatic EVD infection is rare, but paucisymptomatic presentations that fail to trigger fever-based screening cannot be excluded. If the asymptomatic

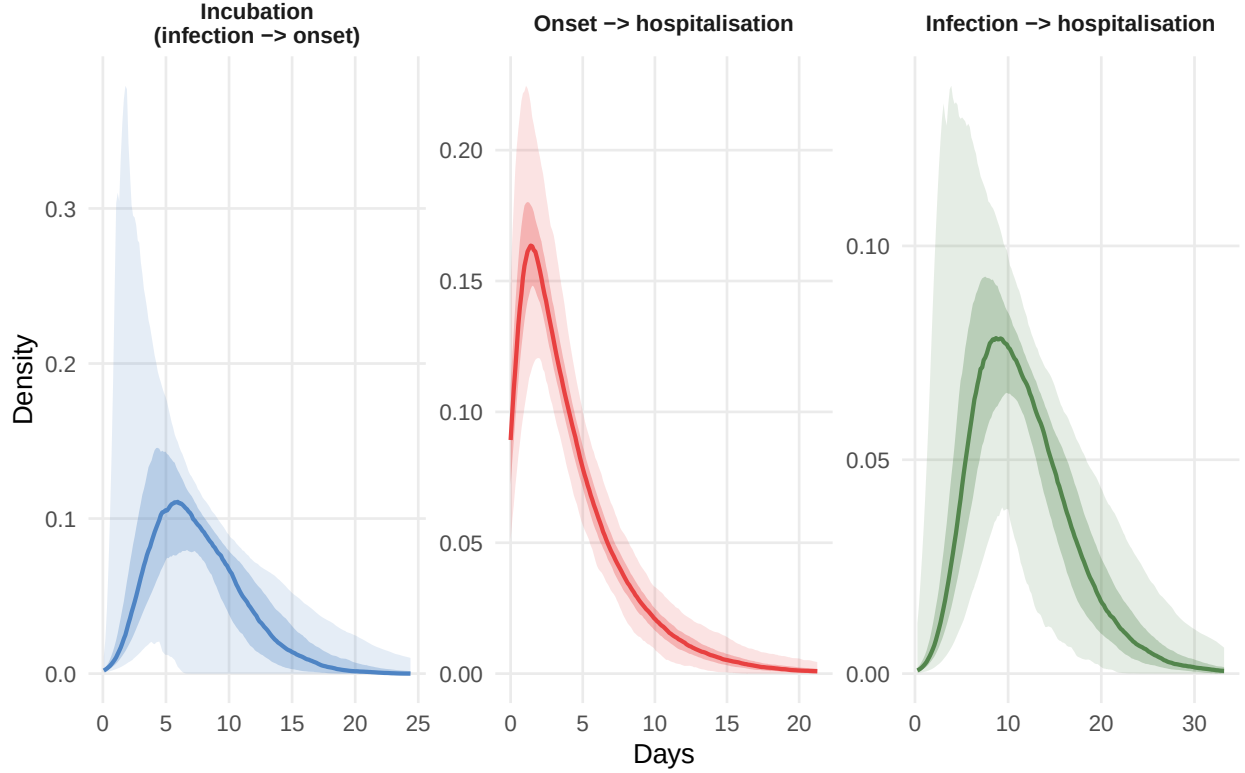


Figure 1: Posterior distributions of natural-history period durations for BDBV. Each panel shows densities from 500 joint draws: the onset-to-hospitalisation delay is drawn from the Bayesian BDBV posterior (Funk & Abbott 2026); the incubation period (infection to symptom onset) is drawn from a Normal prior on the mean (7.68 ± 2.6 days, truncated > 0.5) with CV preserved. Shaded ribbons show 50% (dark) and 95% (light) credible intervals across draws.

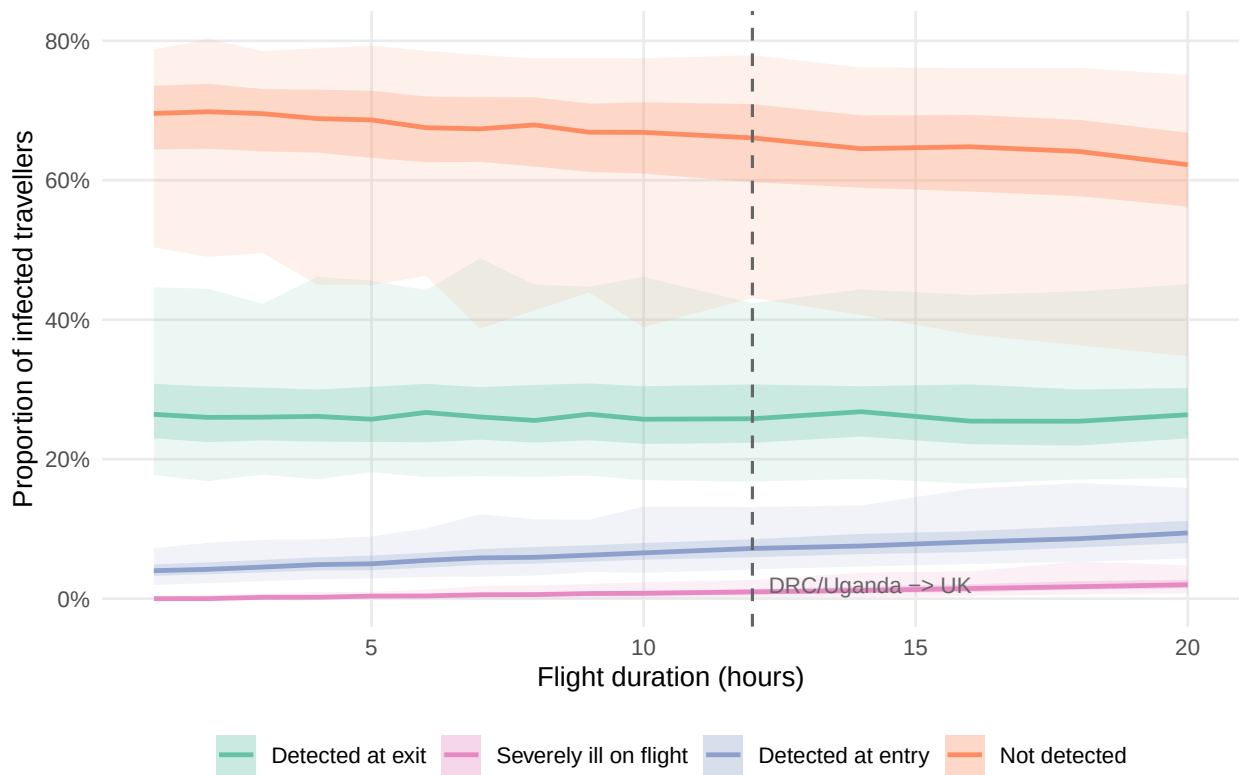


Figure 2: Detection probabilities as a function of flight duration, with uncertainty propagated across 1,000 posterior draws of the BDBV onset-to-hospitalisation delay. Ribbons show 50% (dark) and 95% (light) credible intervals. The vertical dashed line marks 12 hours, the approximate DRC/Uganda-to-UK connecting travel time.

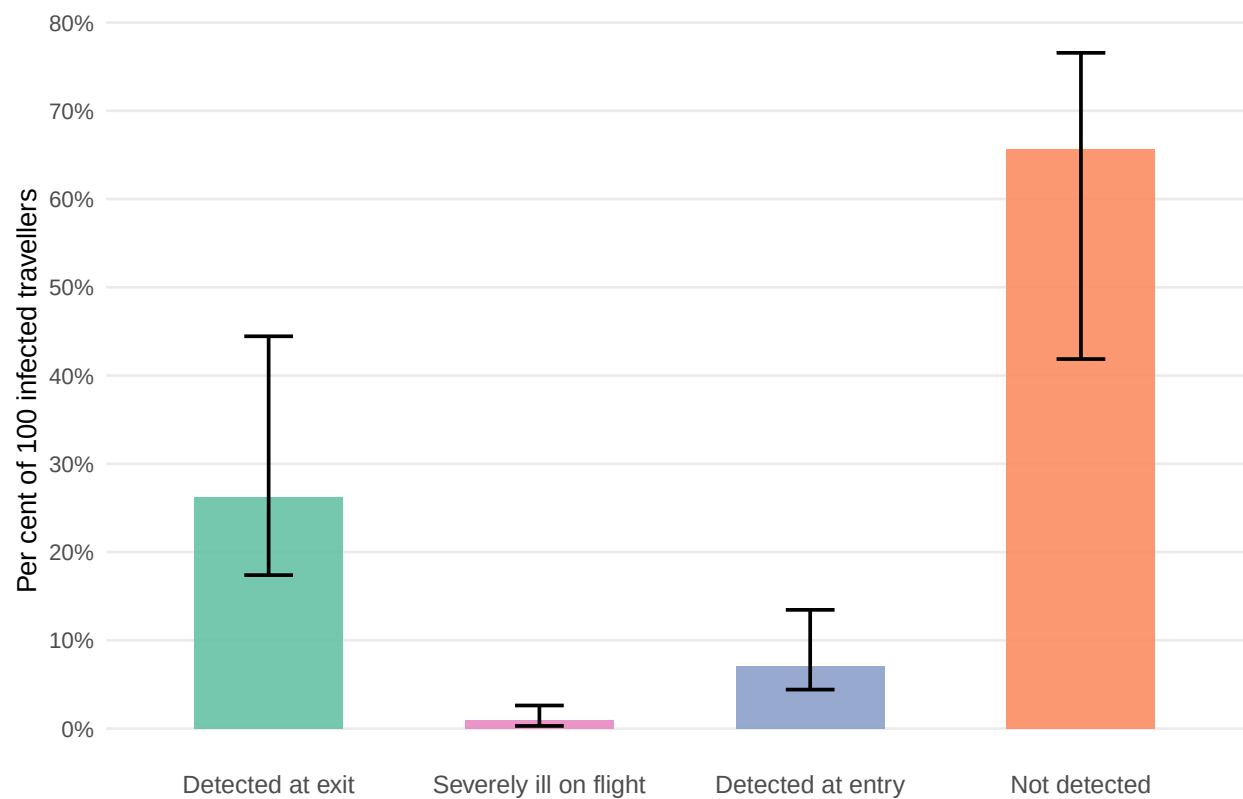


Figure 3: Estimated number of detected and undetected infected travellers per 100 infected travellers departing DRC or Uganda for the UK, stratified by detection outcome, with 95% credible intervals (whiskers).

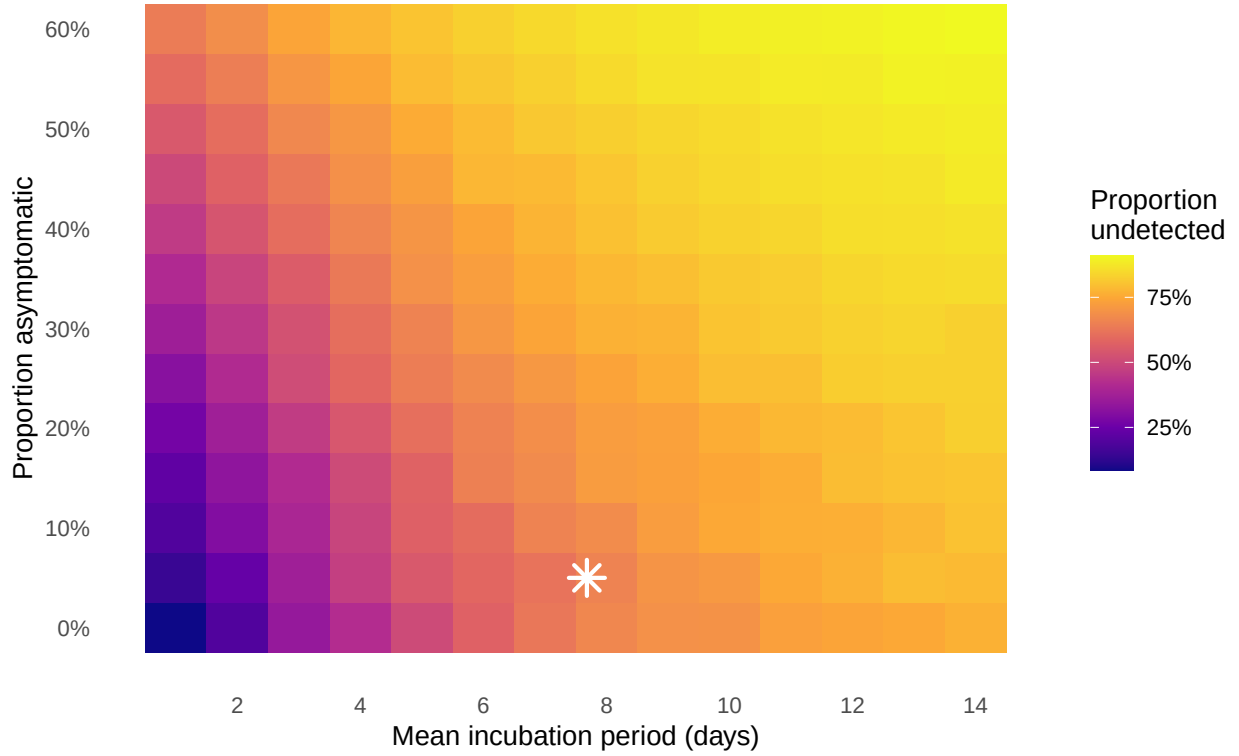


Figure 4: Sensitivity analysis: proportion of infected travellers undetected as a function of the mean incubation period (x-axis) and proportion of asymptomatic infection (y-axis), for a 12-hour flight with posterior-median BDBV onset-to-hospitalisation parameters. The star marks the primary BDBV scenario (mean incubation 7.68 days, 5% asymptomatic). Full uncertainty propagation across the incubation period is shown in the main analysis.

fraction were as high as 40%, the undetected fraction would rise to 46–86% across the same incubation range, but this is considered an unlikely upper bound for BDBV.

3.5 Sensitivity analysis: epidemic growth phase

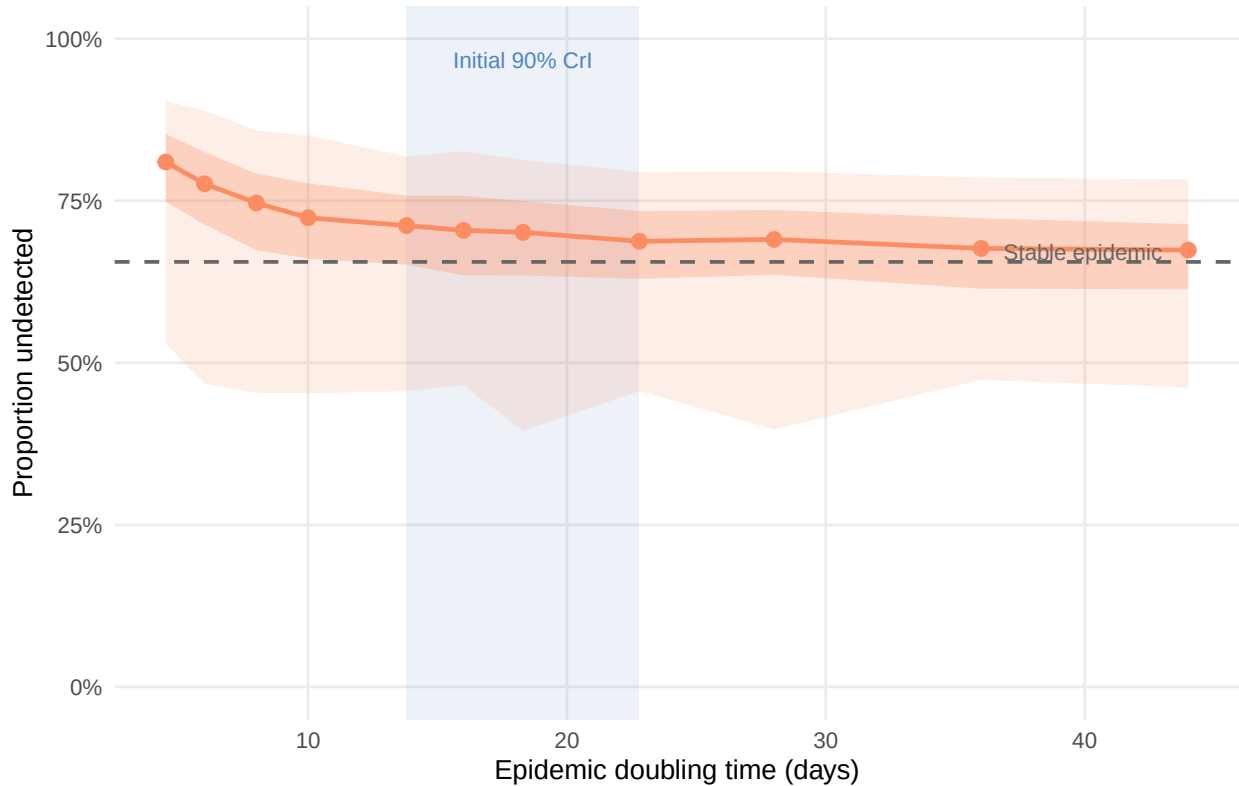


Figure 5: Effect of epidemic doubling time on the proportion of infected travellers going undetected. Each point shows the median across 500 posterior draws; ribbons show 50% (dark) and 95% (light) credible intervals. The shaded band marks the 90% credible interval for the initial doubling time of the current BDBV outbreak (13.8–22.8 days; epiforecasts, 2026). The rightmost marker (dashed vertical) represents a stable epidemic (growth rate = 0; uniform exposure distribution).

The proportion of infected travellers going undetected increases as the epidemic doubling time shortens (Figure 5). Under the initial epiforecasts 90% credible interval for the current BDBV outbreak (doubling time 13.8–22.8 days), the median undetected fraction ranges from approximately 69–71%. Under the more pessimistic latest-estimate lower bound (4.5 days), the undetected fraction rises to 81% (95% CrI: 53–90%), compared with 66% under a stable epidemic. This result highlights that the rapidly evolving phase of an outbreak is precisely the period during which screening is least effective.

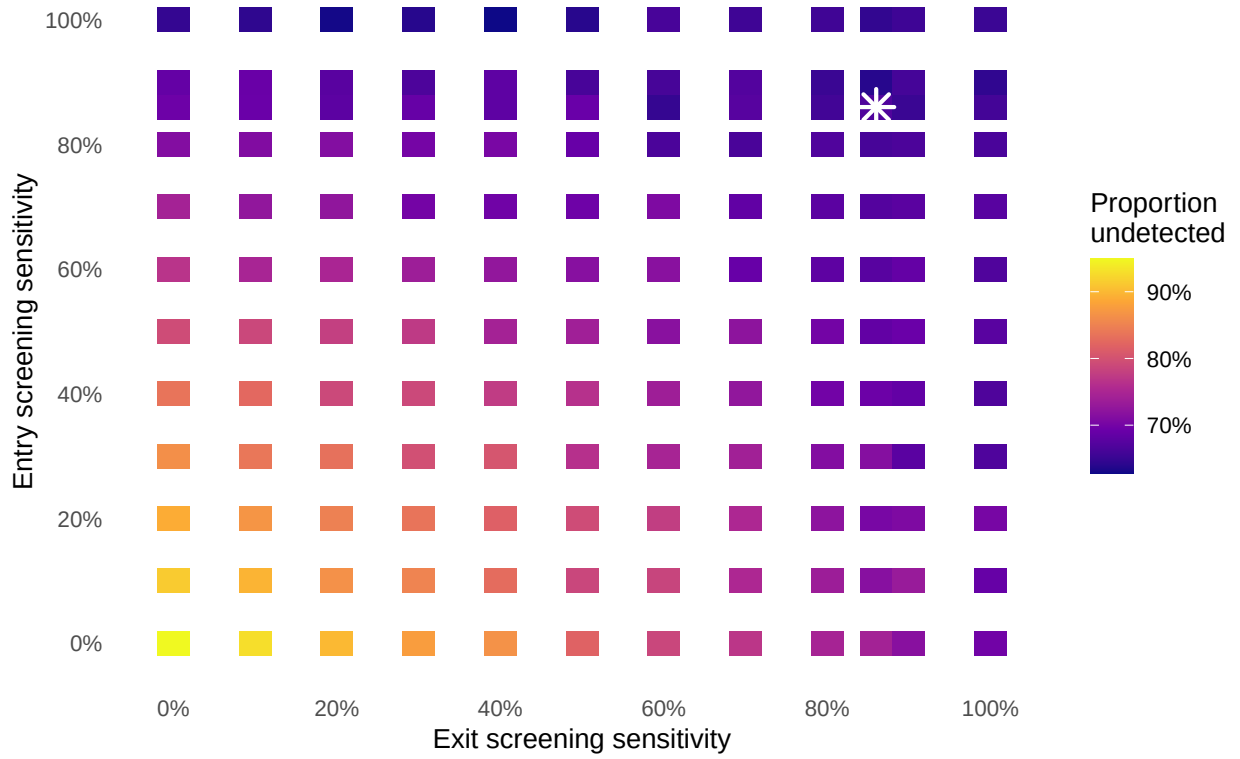


Figure 6: Joint sensitivity analysis: proportion of infected travellers undetected as a function of exit screening sensitivity (x-axis) and entry screening sensitivity (y-axis), for a 12-hour flight with posterior-median BDBV natural-history parameters and stable-epidemic exposure distribution. The star marks the primary scenario (86% at both stages). Diagonal dashed lines indicate contours of equal combined coverage.

3.6 Sensitivity analysis: screening sensitivity at exit and entry

Exit screening sensitivity has a larger marginal effect on the undetected fraction than entry screening sensitivity, because it acts on a larger at-risk window — any traveller who is already symptomatic at departure can be caught at exit, whereas entry screening only adds value for those who become symptomatic during the flight (Figure 6). When exit screening is absent entirely (sensitivity 0%) but entry screening is maintained at 86%, 69% of travellers remain undetected, compared with 74% when entry screening is absent and exit is maintained at 86%. This asymmetry reflects the longer opportunity for symptom detection that the pre-flight period provides relative to the flight itself.

4 Discussion

This analysis estimates that combined exit and entry airport screening for BDBV, at 86% sensitivity and 12-hour DRC-or-Uganda-to-UK travel time, will fail to detect approximately 66% of infected travellers, with a credible interval of 42–77%. This estimate reflects the relatively short onset-to-hospitalisation interval of BDBV — a mean of approximately four days — which means that most travellers who have developed symptoms will progress to detectable severe disease quickly, yet the incubation period (approximately 7.7 days; Wamala et al. 2010) means that many infected travellers will board their flight before they become symptomatic at all.

These findings are broadly consistent with prior modelling of airport screening for Zaire ebolavirus and other viral haemorrhagic fevers. Mabey et al. (2014) (Mabey, Flasche, and Edmunds 2014) estimated that exit screening during the 2014 West Africa ZEBOV epidemic would detect only 7–18% of infected travellers on typical long-haul routes, with combined exit and entry screening performing marginally better; the primary driver of failure was departure during the incubation period. Pitman et al. (2005) (Pitman et al. 2005) showed that even under the assumption of 100% screening sensitivity, entry-only screening for SARS would detect approximately one quarter of infected arrivals, with realistic instrument sensitivity (30–50%) reducing this to fewer than one in ten. Gostic et al. (2015) (K. M. Gostic, Kucharski, and Lloyd-Smith 2015) demonstrated that screening effectiveness is primarily determined by the ratio of the incubation period to the onset-to-severe-disease interval and by the proportion of asymptomatic infection; a subsequent application to COVID-19 estimated that even under best-case assumptions, combined departure and arrival screening would detect a median of only 30% (95% interval 10–53%) of infected travellers in a growing epidemic (K. Gostic et al. 2020). Our BDBV estimates are of a similar order to those for ZEBOV and are somewhat more favourable than the COVID-19 results under low-asymptomatic assumptions, consistent with the relatively short BDBV incubation period and the low observed asymptomatic fraction. Our results show the same qualitative pattern as all prior work: across the plausible range for BDBV, the dominant source of variation in the undetected fraction is the incubation period, with the asymptomatic fraction playing a secondary role given the low asymptomatic rates observed for EVD.

Several important limitations apply to this analysis. First, only three BDBV-specific incubation period estimates exist: Wamala et al. (2010) (mean 7.68 days, $n = 116$) (Wamala et al. 2010), MacNeil et al. (2010) (mean 6.3 days, $n = 24$) (MacNeil et al. 2010), and Kratz et al. (2015) (median 11.3 days, $n = 3$) (Kratz et al. 2015). The Kratz estimate is based on only three cases, and the large range (2–27 days) reflects small-sample uncertainty. We use the Wamala estimate as primary on the basis of the largest sample size and propagate between-study uncertainty in the incubation

prior (SD 2.6 days). Second, the Isiro 2012 outbreak comprised only 52 cases, and the resulting posterior distributions — whilst appropriately wide — are based on a small sample. The Gamma shape and scale posteriors for the onset-to-admission delay have wide 95% credible intervals, and this uncertainty propagates directly into our screening effectiveness estimates. Third, the model assumes a single uniform flight duration of 12 hours; in reality, travellers from DRC and Uganda to the UK take a variety of itineraries with different durations, layovers, and connection points, each of which offers additional opportunities for detection (or re-exposure in a transit lounge). Fourth, the assumed 86% screening sensitivity is derived from a study of thermal image scanning performance during influenza border screening (Priest et al. 2011) and was used in the original 2019-nCoV screening model (Quilty et al. 2020); actual sensitivity for EVD symptom detection in field conditions may differ. Fifth, the model does not account for post-arrival healthcare-seeking behaviour, which represents an important additional detection pathway; infected travellers who develop symptoms after arrival are likely to seek medical care and, if BDBV is considered, to be diagnosed.

Sixth, the simulation model assumes that exposure times are distributed uniformly across the natural history window, which approximates a stable (non-growing) epidemic. In an actively growing outbreak — as the current DRC BDBV situation is likely to be — the distribution of infection ages among travelling cases is skewed towards recent exposure, because recent cases outnumber older ones. Under this scenario, a greater fraction of departing travellers will be pre-symptomatic, and the proportion undetected will be higher than our central estimates suggest. Gostic et al. (2015, 2020) (K. M. Gostic, Kucharski, and Lloyd-Smith 2015; K. Gostic et al. 2020) showed that this epidemic-phase effect can reduce detection, and our growth-phase sensitivity analysis (Figure 5) demonstrates a similar pattern for BDBV: under the initial 90% credible interval for the current outbreak doubling time (13.8–22.8 days; epiforecasts 2026 (Funk and Abbott 2026)), the undetected fraction increases relative to the stable-epidemic baseline, and under the most pessimistic scenario (doubling time 4.5 days) may be higher still. Our central estimates should therefore be interpreted as a lower bound on the undetected fraction during an active growth phase. Seventh, the model does not include questionnaire-based exposure risk screening, which forms part of real-world EVD port-of-entry protocols. Since BDBV transmission requires direct contact with a symptomatic or deceased case, travellers with known exposure history should, in principle, be detectable by structured risk questionnaires even before fever onset. Incorporating this detection pathway — with appropriate assumptions about the fraction of travellers who are aware of their exposure and willing to self-report — would be expected to increase modelled detection rates above the symptom-screening-only estimates presented here.

In summary, airport screening for BDBV is unlikely to detect a meaningful proportion of infected travellers to provide robust containment. Any imported case that escapes detection at both exit and entry represents a potential index case in the UK, underscoring the importance of clinician awareness of BDBV and preparedness to diagnose and isolate suspected cases presenting with febrile illness following travel from affected regions.

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